Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Backend software engineer with 6+ years building production decisioning systems, risk and pricing infrastructure, and distributed services in regulated cloud environments. At Entrata, I built and maintained the eligibility and compliance rule engine used to approve or deny applicants across affordable housing programs (HUD, LIHTC, Section 8) -- the same pattern as underwriting decisioning. Experienced with Python, Java, PostgreSQL, AWS, Kubernetes, Terraform, microservices, and async processing. Currently building ML pipelines and LLM services at Intuit. Eager to learn Go; strong Java and Python foundation accelerates ramp. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / Java / Spring Boot / C# | Go (learning) | PostgreSQL / Advanced SQL | Decisioning & Rule Engine Systems | Distributed Systems / Microservices | Async Processing (RabbitMQ) | AWS (Lambda, S3, RDS, SQS, API Gateway) | Kubernetes | Terraform (IaC) | REST APIs / API Gateway | Caching / Redis | Observability (Prometheus, OpenTelemetry) | ML Pipelines (scikit-learn, XGBoost) | Workflow Orchestration (LangGraph) | CI/CD (GitHub Actions) | Docker

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- Virginia, USA

- Build Python/FastAPI backend services on AWS (Lambda, S3, RDS, SQS, API Gateway); deploy via Docker and GitHub Actions CI/CD; instrument with Prometheus and OpenTelemetry for production observability.
- Build **ML pipelines** (scikit-learn, XGBoost) from feature engineering through prediction APIs -- model evaluation, scoring, and integration with downstream decisioning services.
- Implement input validation, guardrails, and structured-output contracts for LLM services; build golden-dataset regression harnesses to catch quality regressions before release.

Entrata -- Residential & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- Built and maintained the **eligibility decisioning engine** for affordable housing applicant qualification -- AMI threshold calculations, income verification rules, HUD/LIHTC compliance checks, and Approve/Escalate routing across 100K+ units. Zero compliance violations over 6 years.
- Owned **external data integrations**: annual HUD income-limit API ingestion transforming federal source data into ~180,000 domain records, with validation, reconciliation, and rollback at each release cycle.
- Built **RabbitMQ-based distributed async processing** for rent and eligibility updates across thousands of tenant records -- idempotency guarantees, retry logic, and dead-letter handling to prevent double-processing in a multi-tenant system.
- Designed **Docker microservices architecture** (50% better scalability, 30% faster deployments); led AWS S3 migration with PHI encryption (Libsodium) covering 100% of resident data.
- Optimized 200+ SQL queries via execution plan analysis -- 30% performance improvement; wrote complex joins, conditional logic, and stored procedures across a multi-schema PostgreSQL/Oracle environment.
- Built Python/Bash automation reducing manual operational work by 80%; reviewed 300+ PRs; mentored junior engineers on distributed systems and data integrity patterns.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer -- Pune, India

- Built property management backend modules (Python, Java, PostgreSQL) with data import pipelines processing 1.6M HUD/Tax Credit records; implemented role-based access control and multi-tenant data isolation.

PROJECTS

Leasel -- Compliance Decisioning System

Python · LangGraph · FastAPI · Workflow Orchestration · PostgreSQL

8-agent LangGraph workflow orchestrating decisioning for housing compliance: triage, data retrieval, rule evaluation, confidence scoring, and human escalation at defined thresholds. Each agent has a clear goal, tool set, verification step, and stop condition -- the same loop engineering pattern used in underwriting systems. LLM-as-judge CI: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- ML Inference & Evaluation Infrastructure

Python · FastAPI · Provider Routing · CI/CD · Observability

15-endpoint FastAPI service with provider routing across five LLM/ML inference backends. Scoring engine evaluates output quality on five metrics; CI regression against golden datasets. Same pattern as ML infrastructure for underwriting model evaluation.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Clinical Decisioning API

Python · Flask · REST APIs · RBAC · Docker · HIPAA

RESTful decisioning API translating clinical queries into FHIR data calls with fine-grained authorization. Role-based access control, HIPAA-style data isolation, and containerized deployment -- regulated-data decisioning pattern.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017